	GLOBAL ECONOMICS **Grade 10** **Catch ups exam** **Decision-making under uncertainty** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Characterise economic decisions under uncertainty, organise payoff tables, calculate profits from revenues and costs, and justify choices using Maximax, Maximin, Minimax Regret, EMV/Bayes, and probability sensitivity analysis.	**Assessment criteria** **C1.** Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision. **C2.** Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison. **C3.** Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison. **C4.** Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations. **C5.** Constructs opportunity loss tables and applies the Minimax Regret criterion to select the alternative that minimises the maximum opportunity loss. **C6.** Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff. **C7.** Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

This exam assesses C1–C7. Each question and task shows its points. **Passing requires a minimum of 5 cumulative points per criterion, including points earned in previous learning activities**. Show all procedures; answers without the required procedure receive no credit. Partial credit may be awarded for a correct method with arithmetic errors. Missing names or late submissions incur a 1-point deduction per criterion.

№	C1	C2	C3	C4	№	C5	C6	C7
1	3	6	–	–	6	2	–	–
2	3	–	6	–	7	2	–	–
3	–	–	–	2	8	2	–	–
4	–	–	–	2	9	–	2	–
5	–	–	–	2	10	–	2	–
					11	–	2	–
					12	–	–	3
					13	–	–	3
Tot.	**6**	**6**	**6**	**6**	**Tot.**	**6**	**6**	**6**

1. Community Science Sessions [C1-3 C2-6]

A community science centre must choose an astronomy session or a robotics session before family demand is known. Demand will be busy or quiet. Astronomy provides a net contribution of 14 USD per participant and a fixed sponsorship contribution of 90 USD per session. Robotics provides 17 USD per participant and a fixed sponsorship contribution of 65 USD per session. Busy demand adds a further net contribution of 5 USD per participant to either session; quiet demand adds 3 USD per participant. Astronomy attracts 28 participants with busy demand or 17 with quiet demand; robotics attracts 23 or 14, respectively. All contributions are additive, and no other adjustments are needed.

Characterize the key elements of the decision-making situation **[C1-3]** and construct the payoff table. **[C2-6]**

2. Kayak Rentals [C1-3 C3-6]

A kayak rental business must choose single kayaks or tandem kayaks before seasonal demand is known. Demand will be busy or quiet. Single kayaks generate 84 rentals with busy demand or 56 with quiet demand; tandem kayaks generate 66 or 38, respectively. Base revenue per rental is 36 USD for the single kayaks and 49 USD for the tandem kayaks. Busy demand adds a surcharge of 7 USD per rental to either rental type; quiet demand adds 4 USD per rental. Variable cost per rental is 20 USD for the single kayaks and 29 USD for the tandem kayaks. Total fixed seasonal costs, for either rental type, are 540 USD with busy demand and 620 USD with quiet demand. These fixed costs are paid once per season, and there are no other revenues or costs. Characterize the key elements of the decision-making situation **[C1-3]** and construct the payoff table. **[C3-6]**

3. Bicycle Assembly [C4-2]

A bicycle manufacturer must select an assembly plan before market conditions are known. Licensed production earns a fixed royalty regardless of the market state. The possible states, in column order, are Boom, Stable market, Recession. Profits are in thousand USD. No probabilities are available.

Alternative	S_1	S_2	S_3
Large factory	132	53	-27
Standard line	91	63	19
Compact line	61	47	33
Licensed production	26	26	26

What would the optimistic decision be? What would the conservative decision be? Justify your answer. **[2]**

4. Solar Installation [C4-2]

A solar installer must select an installation programme before customer adoption is known. The service agreement pays a fixed profit under every adoption state. The possible states, in column order, are Rapid adoption, Moderate adoption, Slow adoption. Profits are in thousand USD. No probabilities are available.

Alternative	S_1	S_2	S_3
Regional rollout	168	51	-39
Town rollout	108	69	17
Village rollout	73	52	31
Service agreement	35	35	35

What would the optimistic decision be? What would the conservative decision be? Justify your answer. **[2]**

5. Equipment Leasing [C4-2]

A construction-equipment business must choose a fleet plan before next season's rental utilisation is known. The possible states, in column order, are Peak utilisation, Busy utilisation, Routine utilisation, Quiet utilisation. Profits are in thousand USD. No probabilities are available.

Alternative	S1	S2	S3	S4
Buy large fleet	156	81	21	-46
Buy medium fleet	113	85	41	11
Lease small fleet	77	63	49	25
Pool equipment	54	51	39	32

What would the optimistic decision be? What would the conservative decision be? Justify your answer. **[2]**

6. Festival Meals **[C5-2]**

A festival caterer must select a meal-preparation plan before attendance is known. The possible states, in column order, are Crowded, Average attendance, Sparse attendance. Profits are in thousand USD. No probabilities are available. All payoffs are to be maximised.

Alternative	S_1	S_2	S_3
Full menu	87	37	-16
Balanced menu	65	54	21
Limited menu	43	41	34

What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

7. Laundry Capacity **[C5-2]**

A commercial laundry must select its capacity plan before hotel orders are known. A contracted service provides a fixed profit under every workload state. The possible states, in column order, are Heavy workload, Routine workload, Light workload. Profits are in thousand USD. No probabilities are available. All payoffs are to be maximised.

Alternative	S_1	S_2	S_3
Build plant	126	47	-32
Upgrade machines	94	73	19
Lease machines	69	61	38
Contract service	41	41	41

What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

8. Juice Production **[C5-2]**

A juice producer must choose its daily production plan before sales are known. The possible states, in column order, are Exceptional sales, Brisk sales, Ordinary sales, Poor sales. Profits are in USD. No probabilities are available. All payoffs are to be maximised.

Alternative	S1	S2	S3	S4
Large batch	1080	610	230	-270
Staged batches	810	730	510	150
Small batch	530	550	480	390

What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

9. Coffee Roasting **[C6-2]**

A coffee roaster must choose a roasting plan before export conditions are known. The possible states, in column order, are Export boom, Regular exports, Export slump. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
$P(S_j)$	0.35	0.40	0.25
Specialty roast	98	49	-17
Blended roast	74	63	27
Everyday roast	51	53	42

What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

10. Farm Visitor Facilities **[C6-2]**

A farm must select visitor facilities before seasonal attendance is known. Leasing to an operator earns a fixed profit under all attendance states. The possible states, in column order, are Busy season, Typical season, Quiet season. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
$P(S_j)$	0.25	0.45	0.30
Visitor centre	117	43	-29
Covered terrace	89	64	23
Picnic shelter	63	57	40
Lease to operator	32	32	32

What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

11. Translation Capacity **[C6-2]**

A translation agency must choose a capacity plan before client orders are known. The possible states, in column order, are Order rush, Busy orders, Routine orders, Few orders. Profits are in thousand USD.

Alternative	S1	S2	S3	S4
$P(S_j)$	0.15	0.35	0.25	0.25
Permanent staff	137	81	29	-43
Staff plus freelancers	99	86	59	17
Core staff	65	67	61	48

What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

12. Mobile Game Release **[C7-3]**

A game developer must choose a wide release or a limited release before player uptake is known. The possible states, in column order, are Viral uptake, Moderate uptake, Low uptake. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
Wide release (A_1)	108	44	-10
Limited release (A_2)	58	64	30

Suppose $P(S_1) = p$ and $P(S_2) = p/2$; determine $P(S_3)$ from the sum-to-one condition.

State the preferred alternative for every probability value ($0 \leq p \leq 1$). **[3]**

13. Timber Supply Agreement **[C7-3]**

A timber merchant must choose spot sales or a reserved-supply agreement before next season's timber availability is known. The possible states, in column order, are Scarce timber, Balanced supply, Abundant timber. Profits are in thousand USD.

Alternative	S_1	S_2	S_3
Spot sales (A_1)	30	66	94
Reserved supply (A_2)	85	46	40

Suppose $P(S_1) = 2p$ and $P(S_2) = p$; determine $P(S_3)$ from the sum-to-one condition.

State the preferred alternative for every probability value ($0 \leq p \leq 1$).. **[3]**